

Assessment three: Spatial Analysis Workflow Project

Project Topic: What is the best location in Warkworth for a new bus station when the northern expressway bus service extends further north?

Introduction

Bus stops are essential for ensuring reliable and organized public transit, making daily travel more convenient and accessible for everyone. As urban areas expand and regional transport networks evolve, the strategic location of transport infrastructure such as bus stations plays a crucial role in shaping accessibility, mobility, and urban growth. With the planned extension of the Northern Expressway bus service further north, Warkworth, a rapidly growing township north of Auckland, becomes a key node in the regional transit network. Identifying the most suitable location for a new bus station in Warkworth requires a spatially informed decision-making process that considers multiple geographic, social, and infrastructural variables.

In the context of regional towns like Warkworth, where infrastructure is developing in tandem with population growth, the integration of bus services with existing and planned urban features becomes even more critical. According to Currie and Delbosc (2011), stations placed in areas with high walkability, proximity to key services, and seamless road connectivity tend to see higher ridership and community support. This project uses a GIS-based approach to assess land suitability through a structured spatial workflow grounded in academic research and local planning policies.

Background Research

Identifying a suitable location for a new bus station in Warkworth, an expanding township at the northern edge of Auckland's metropolitan reach requires a comprehensive spatial analysis that integrates infrastructure, landform, accessibility, and population data. With the anticipated extension of the northern expressway bus service, Warkworth is poised to become a strategic node in Auckland's wider public transport network. Selecting the most appropriate site for a new bus station is not only a matter of spatial convenience but also one of long-term urban sustainability, accessibility, and safety.

Public transport infrastructure siting is a well-studied topic in geographic and urban planning literature. Various international and national studies emphasize the importance of a multi-criteria approach considering topography, land use, accessibility, population demand, and hazard zones such as flood plains (Farahani et al., 2013; Bhaskar & Chung, 2013). In New Zealand, such approaches are increasingly used by local councils and transport agencies to ensure that investments into infrastructure are strategically aligned with future growth patterns and social equity considerations (Auckland Council, 2020).

Importance of Location and Accessibility in Transit Planning

Accessibility plays a central role in public transport infrastructure planning. A well-located bus station enhances mobility for a wider population, improves modal interchange efficiency, and reduces reliance on private vehicles. Studies from Auckland Transport (2019) show that areas with high walkability and proximity to schools, health services, and commercial zones have higher public transport uptake. Placing a station near major road corridors increases operational efficiency and reduces construction costs, while simultaneously enabling easier integration with existing bus routes and stops. In this project, NZ road centrelines and GTFS bus stop data were used to determine how closely potential sites align with existing transport infrastructure, ensuring minimal disruption and maximum connectivity.

The effectiveness of a transport hub also depends on the surrounding land use. Areas with mixed-use zones such as residential, commercial, and institutional are more likely to generate steady foot traffic throughout the day (Curtis & Scheurer, 2016). Therefore, land use suitability becomes a fundamental layer in spatial analysis. OpenStreetMap (OSM) data, a reputable crowdsourced geospatial platform, offers classified land use layers that are particularly useful in urban-scale planning projects like this.

Topography and Suitability

Topographic constraints are often overlooked in urban planning until engineering assessments begin. However, in spatial decision-making, slope is a primary consideration, especially for infrastructure that must be universally accessible. A site with steep slopes introduces construction complexity, increases cost, and impedes mobility for elderly or disabled users. In this project, a Digital Elevation Model (DEM) with 8-metre resolution from LINZ was used to generate a slope suitability map. As recommended in accessibility-focused design literature, sites with slopes less than 5 degrees were considered suitable (Krieger & Higgins, 2002). The slope suitability map formed one of the key inputs in the weighted overlay analysis to prioritise ease of access and construction viability.

Flood Risk and Infrastructure Resilience

Climate resilience is a growing concern in infrastructure planning. In New Zealand, localised flooding events have become more frequent due to climate change and urban runoff (Ministry for the Environment, 2020). Auckland Council provides spatial datasets for floodplains and flood risk zones, and these were used in this project to exclude unsuitable areas. Incorporating flood risk ensures that the proposed infrastructure investment is long-lasting and avoids future disruptions, repair costs, or safety concerns. This analysis is especially important for Warkworth, where parts of the township lie within low-lying terrain susceptible to water accumulation during storms.

Population Demand

Public transport must serve people, so understanding where people live is fundamental to deciding where to place transport facilities. Using Stats NZ meshblock population data from the

latest census, the project identified population meshblocks that are likely to contribute to consistent usage of the new bus service. This analysis considered population density alongside walkability, showing areas where public transit demand is naturally concentrated. The importance of integrating population demand into spatial models is reinforced in studies such as Currie and Delbosc (2011), who highlight that demand-driven planning can improve both cost-effectiveness and equity of service provision.

The Role of Land Use

The effectiveness of a transport hub also depends on the surrounding land use. Areas with mixed-use zones such as residential, commercial, and institutional are more likely to generate steady foot traffic throughout the day (Curtis & Scheurer, 2016). Therefore, land use suitability becomes a fundamental layer in spatial analysis. OpenStreetMap (OSM) data, a reputable crowdsourced geospatial platform, offers classified land use layers that are particularly useful in urban-scale planning projects like this.

Spatial Methods in Bus Station Siting

The methodology used in this project aligns with global best practices in Geographic Information System (GIS)-based suitability analysis. A multi-criteria decision analysis (MCDA) framework was implemented within QGIS, combining raster and vector spatial datasets using reclassification, buffering, and weighted overlay. Each factor such as slope, population, proximity to roads was rasterised and normalized to a common scale, allowing for logical combination. This method is recommended in many spatial planning case studies for its transparency, flexibility, and reproducibility (Malczewski, 2006).

GTFS (General Transit Feed Specification) datasets were used to map existing routes and stops. These helped identify areas already serviced by public transport and those needing expansion. Auckland Transport and GTFS feeds have become industry standards for transport network modelling across New Zealand and internationally, due to their rich, standardised, and easily integrable format (NZTA, 2023).

Precedents and Local Context

Auckland's Unitary Plan and recent transport development initiatives show an increasing emphasis on transit-oriented development (TOD). Similar projects, such as the upgrade of the Hibiscus Coast Busway Station, demonstrate the importance of placing new infrastructure near key arterial roads and within growth zones designated by the Auckland Unitary Plan (Auckland Transport, 2021). Warkworth is specifically marked as a future urban growth area, making this project timely and necessary. In regional contexts like Warkworth, where urban development meets rural fringes, ensuring equitable access is critical. The use of walk and cycleway datasets supports active transport integration, aligning with national transport strategies that aim to reduce carbon emissions and promote healthy mobility (Ministry of Transport, 2021).

This background review confirms that a spatially driven, multi-criteria approach is both academically and practically justified for determining the best location for a bus station in Warkworth. By incorporating local terrain, land use, accessibility, population needs, and other factors this project builds upon current best practices in transport infrastructure planning and aligns with both national strategy and community needs.

Scope of the Project

This project aims to determine the most suitable location for a new bus station in Warkworth, a rapidly growing township north of Auckland, New Zealand. With the planned extension of the Northern Expressway (NX1) service, the need for a strategically located bus terminal becomes critical to enhance connectivity between Warkworth and the wider Auckland metropolitan area. A spatially driven, evidence-based approach has been used to identify the optimal location, balancing accessibility, demand, land suitability, and environmental resilience.

Geographic Scope

The spatial analysis is confined to the urban and semi-urban extents of Warkworth, mapped using the New Zealand Transverse Mercator 2000 (EPSG:2193) coordinate reference system. The study area covers built-up and developable areas based on OpenStreetMap (OSM) land use classifications, focusing on residential, commercial, and mixed-use areas. Rural zones were excluded unless they demonstrated high strategic value in terms of connectivity and future development potential.

The project boundary was defined through visual interpretation of aerial basemaps, OSM features, and contextual overlays such as road centrelines and walk/cycle infrastructure. Meshblock population data helped establish the spatial pattern of likely service demand across Warkworth.

Thematic Scope

This project assesses site suitability for a future bus station by combining multiple geospatial criteria in a structured spatial analysis workflow. The key factors considered include:

- Accessibility: Proximity to existing roads, walk/cycleways, and GTFS bus stops
- Population demand: Concentration of residents and proximity to community facilities
- Environmental constraints: Exclusion of flood-prone areas and land with slopes above 5°
- Land use compatibility: Based on OSM land use, prioritising non-residential zones where development is feasible

These thematic layers were prepared using QGIS tools including reclassification, buffering, raster conversion, and weighted overlay. Suitability was determined using a composite raster that integrates all factors with assigned weights based on relevance and planning logic.

Assumptions

Several assumptions guided the spatial modelling and helped ensure alignment with best practices in transport and urban planning:

- Sustainability and accessibility are the highest priorities, reflecting the long-term vision for Warkworth's transport infrastructure.
- Sites near main roads and GTFS bus stops are considered more viable due to ease of service integration.
- Locations within flood-prone areas or on steep terrain (slope $>5^\circ$) are considered unsuitable for heavy public infrastructure.
- High walkability and cycling access increase station usefulness for the local community.
- Mixed-use and non-residential land types, as mapped from OSM, offer greater flexibility for public infrastructure development.

These assumptions are consistent with urban planning literature that emphasises connectivity, climate resilience, and multimodal transport integration (Curtis & Scheurer, 2016). The final output is a spatially explicit suitability model supporting informed, future-ready decision-making for public transport infrastructure.

Identify Data Needs

To conduct a robust spatial analysis, we identified and acquired 10 core datasets from various authoritative and open sources. These datasets are processed in QGIS 3.42 using geospatial workflows, including buffering, rasterization, and overlay analysis. Following is the details of used datasets in the project:

Sl.	Dataset	Format	Source	Type	Description
01	Digital Elevation Model (DEM) 2012	GeoTIFF	LINZ Data Service	Raster	Used to calculate terrain slope. Steep areas are avoided for station infrastructure.
02	Population (Meshblocks 2018)	Shapefile	Stats NZ	Vector	Used to identify areas with highest population concentration and predict demand.
03	Road Centrelines	Shapefile	LINZ Data Service	Vector	Used to calculate proximity and access to the transport network.
04	Land Use (OSM)	GeoPackage	OpenStreetMap NZ Extract	Vector	Used to classify zones: commercial, residential, industrial.
05	Bus Stops	GeoJSON	Auckland Transport GTFS	Vector	Used to understand existing public transport network and connectivity.

06	Bus Routes	GeoJSON	Auckland Transport GTFS	Vector	Helps contextualise current and potential bus corridors.
07	Walk & Cycleways	Shapefile	Auckland Council Data	Vector	Accessibility factor for sustainable commuting.
08	NZ Facilities	Shapefile	Koordinates	Vector	Indicates areas of daily human activity to ensure high demand zones (Hospitals, Schools).
09	Flood Plains	Shapefile	Auckland Council Flood Map	Vector	Areas with flood risk were excluded due to infrastructure vulnerability.
10	OSM Basemap	Web Layer	OpenStreetMap QGIS plugin	WMS	Used for background map and spatial orientation.

Data Quality & Implications

All datasets were projected to EPSG:2193 to ensure spatial consistency. Raster resolution was standardised at 8x8 metres, balancing detail and processing efficiency.

Potential limitations:

- DEM is from 2012 and may not reflect recent topographical changes.
- Crowdsourced OSM land use may have inconsistencies in classification.
- GTFS data reflects current routes only; future route extensions are speculative.

Despite these, the dataset ensemble provides a sufficiently accurate spatial approximation for decision-making.

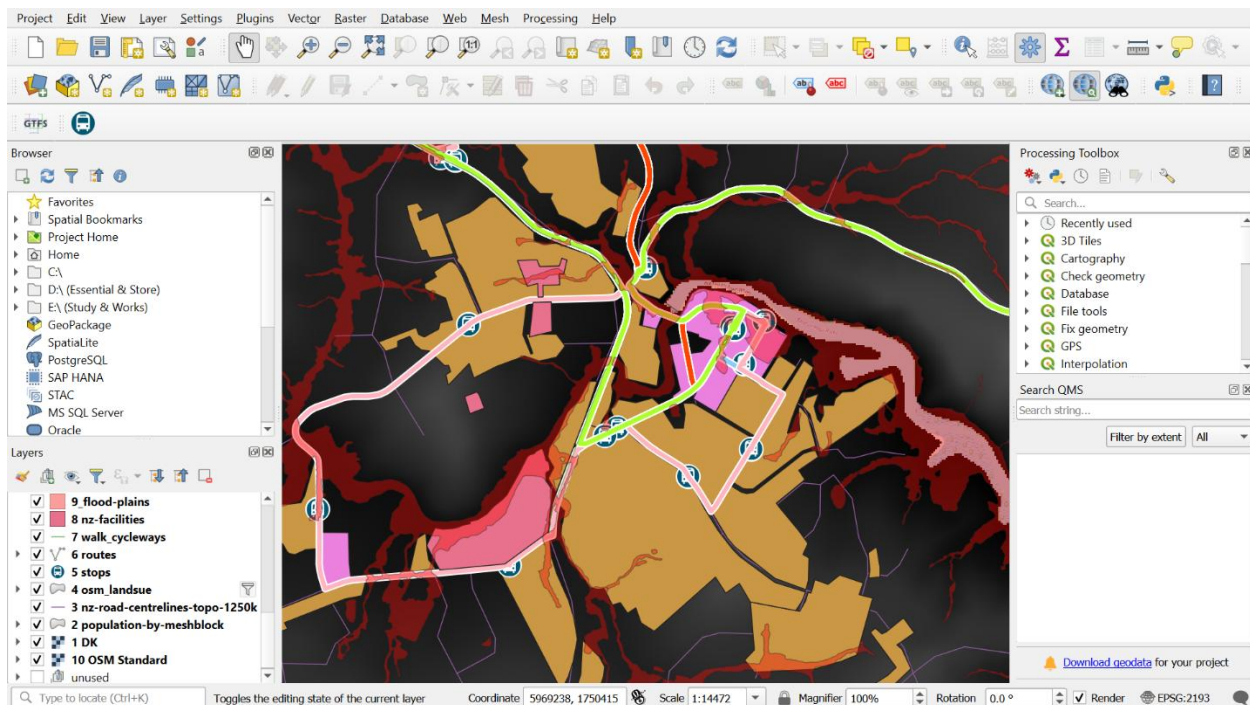
Privacy & Ethical Considerations

- Meshblock data is aggregated and anonymised, mitigating privacy risks.
- No personally identifiable or sensitive data was used.
- Open data licenses such as LINZ, Stats NZ Open License were adhered to.
- Ethical mapping principles were followed, particularly in excluding flood-risk zones to avoid recommending unsafe locations.

Loading Datasets

- Vector Layers: Layer => Add Layer => Add Vector Layer
- Raster Layers: Layer => Add Layer => Add Raster Layer
- Web Plugins (installed):
 - Web => QuickMapServices => OSM => OSM Standard (for basemap)
 - Web => GTFS Go => GTFS Go (For Bus Stops and Routes)

The screenshot below shows the initial data layers successfully loaded into the QGIS environment, illustrating the multi-layered spatial framework used for subsequent analysis.



Map 1: Loaded all datasets

Perform Analysis

To determine the most suitable location for a new bus station in Warkworth, a spatial multicriteria analysis (MCA) approach was conducted using QGIS 3.42.0. This approach integrates multiple geospatial datasets, transforms them into comparable formats, and evaluates them using weighted overlay techniques. MCA has been widely applied in similar location-allocation problems for transport infrastructure (Malczewski, 2006). The goal was to spatially quantify factors like accessibility, safety, environmental suitability, and population proximity using a combination of raster and vector geoprocessing tools.

Phase 1: Slope Analysis (Raster)

Slope is a critical factor in transport planning, as flat terrain supports infrastructure development and accessibility (Nasiri et al., 2018).

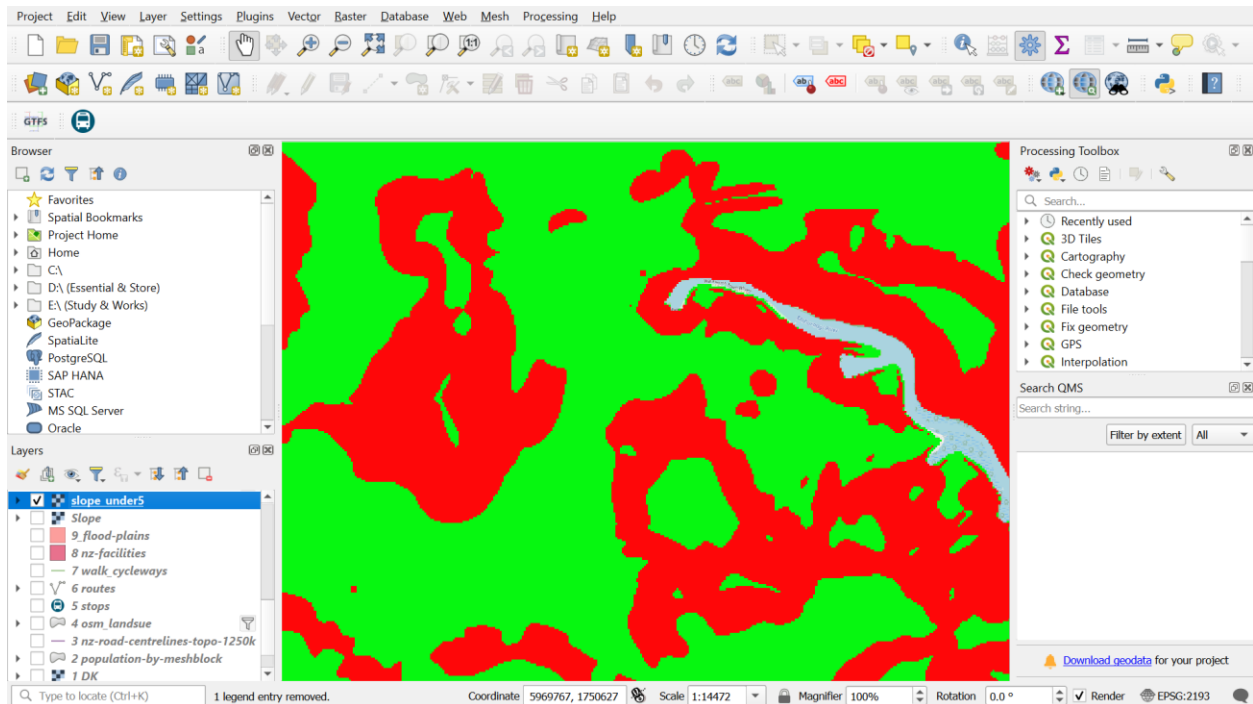
The 2012 Digital Elevation Model (DEM) was reclassified to isolate flat areas where slope is under 5°.

Process:

- Raster => Analysis => Slope
- Raster Calculator: Used expression "slope@1" < 5 to produce binary raster.

- Output: Areas under 5° slope = 1 (suitable), rest = 0.
- Style output: green for suitable, red for unsuitable.

Following is the view after styling the output, green areas indicate suitable locations, while red areas refer to the unsuitable areas.



Map 2: Slope suitability

Phase 2: Accessibility – Proximity Buffers

Accessibility to existing infrastructure is crucial. I buffered features that enhance accessibility or services: roads, facilities, bus stops, routes, and walk/cycleways.

Datasets:

- NZ Road Centrelines
- GTFS Bus Stops
- NZ Facilities
- Walk and Cycleways

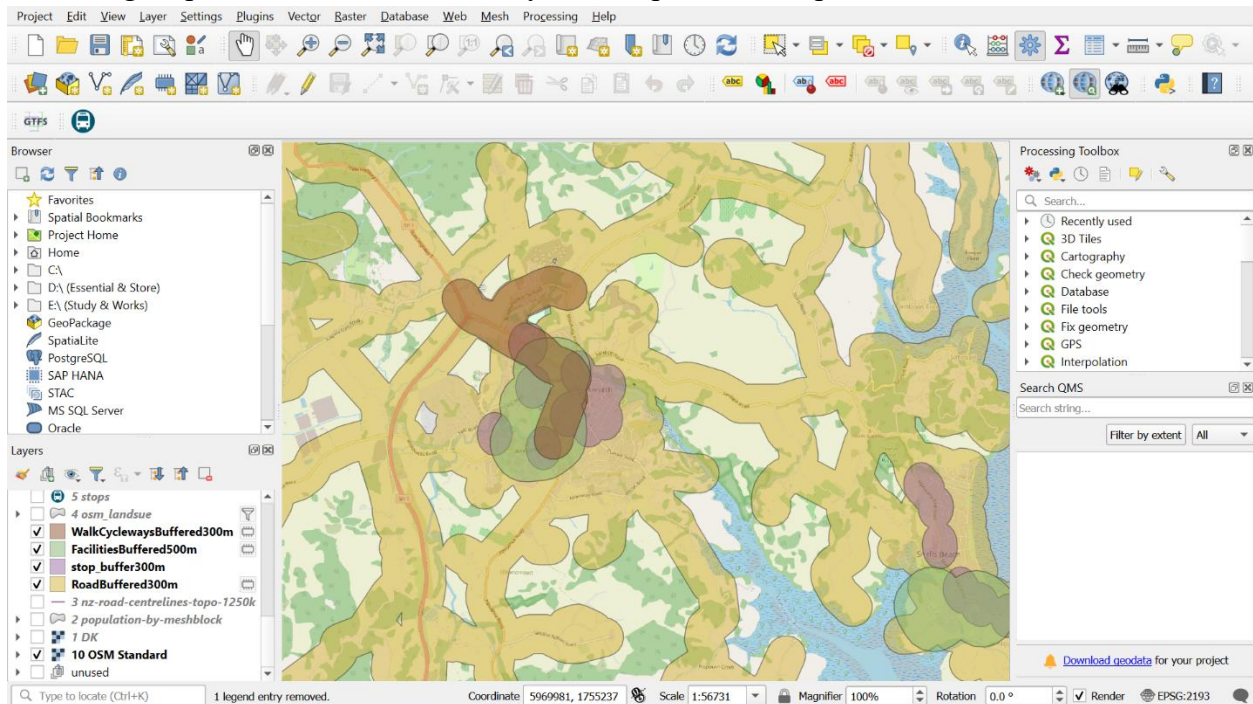
Buffer Distances were measured based on various urban transport planning studies:

- Roads: 300 m
- Bus stops: 300 m
- Facilities: 500 m
- Walk/Cycleways: 300 m

Process:

- Vector => Geoprocessing Tools => Buffer
- CRS: EPSG:2193 (NZTM)
- Each buffered vector layer was saved and visualized.
- Symbology opacity was set at 50% for each buffered layer to clarify visualization.

Following map illustrates the buffered layers on OpenStreetMap base for better visualization:



Map 3: Proximity Buffers

Phase 3: Rasterization of Vector Layers

In this phase all the layers are converted into raster layer. Raster data allows arithmetic overlay operations essential for multicriteria analysis.

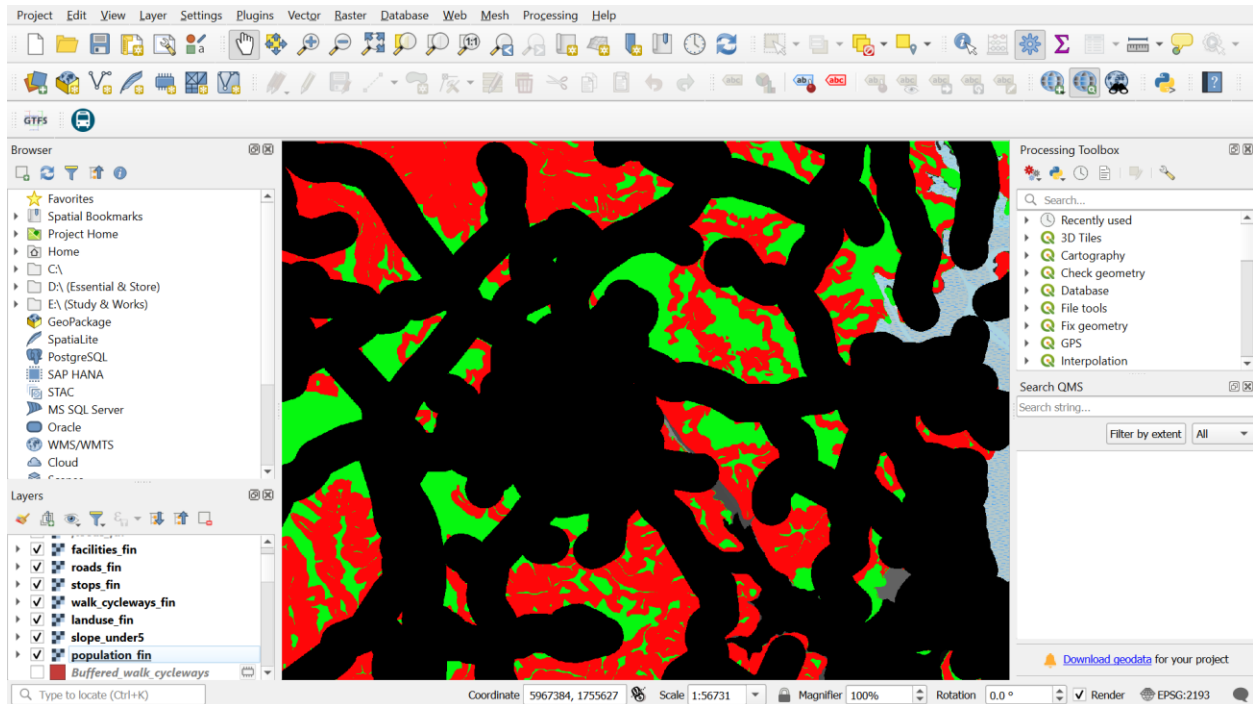
Legends:

- NZ Facilities
- Road Centrelines
- Bus Stops
- Walk Cycleways
- Land use
- Slope
- Population meshblock

Process:

- Vector => Raster => Rasterize (vector to raster)
- Output raster size unit: Georeferenced units
- Pixel size: 8x8 (to align with slope layer)
- CRS: 2193 for consistency

Following is the view in QGIS of all rasterized layer found at this stage:



Map 4: After rasterized layers

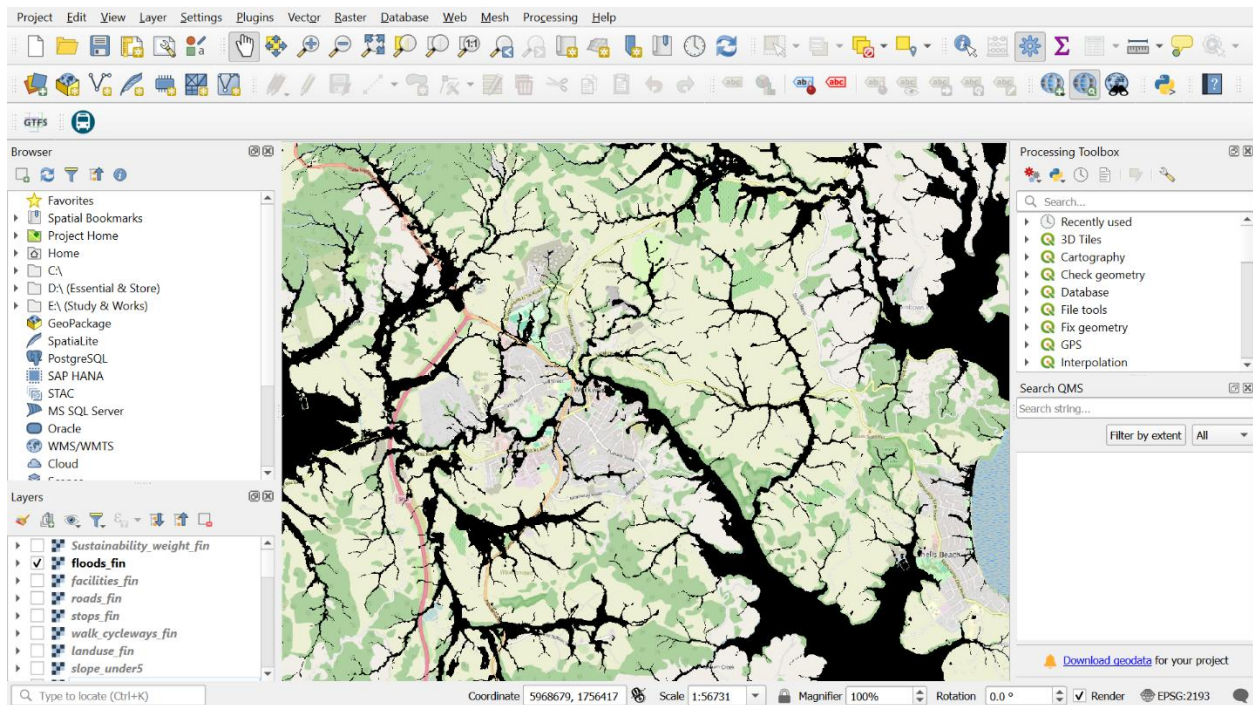
Phase 4: Prepare Flood Plain to Exclude (Environmental Constraints)

Flood-prone areas are unsuitable due to infrastructure vulnerability and public safety risks (Auckland Council, 2017).

Method:

- Convert flood polygon to raster (value = 1)
- Raster Calculator used to subtract flood areas from overall suitability (next phase).

Following map indicates the environmental constraint- flood plains to exclude during raster calculation:



Map 5: Flood pains to exclude

Phase 5: Final Suitability Calculation - Weighted Raster Overlay

Weighting allows prioritization based on significance. To produce a composite raster output reflecting the most suitable areas for a new bus station in Warkworth based on multiple spatial criteria. Each raster input was assigned a weight representing its relative importance in influencing the station's optimal location.

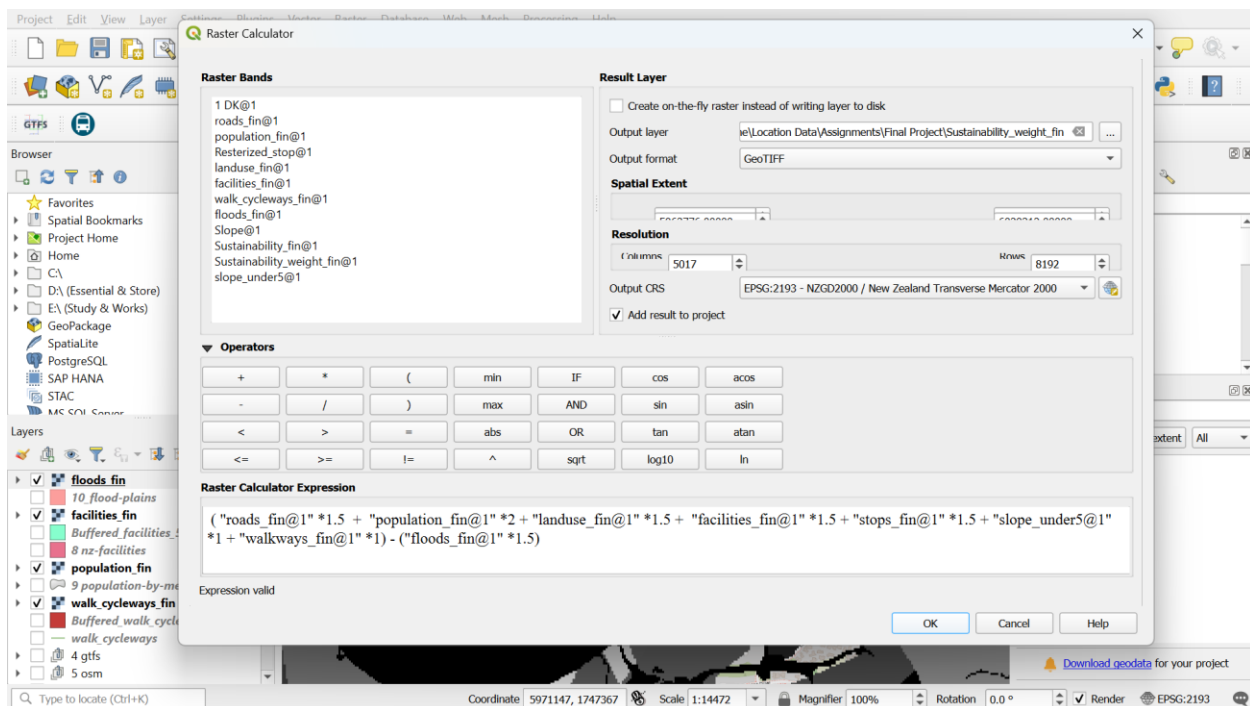
Weights Assigned:

- **Population: 20%.** This is the most important factor. A higher population density increases ridership potential. Prioritizing this ensures the bus station serves the maximum number of people.
- **Roads: 15%.** This is critical for connectivity. Road access is essential for both buses and users arriving on foot or by vehicle. NZ Transport Agency guidelines emphasize the importance of bus stations being located along major road corridors
- **Land Use: 15%.** Land use patterns directly affect service demand and feasibility. Residential and mixed-use zones provide a ready catchment for riders and offer complementary amenities.
- **Facilities: 15%.** Public facilities such as, schools, hospitals generate regular traffic and increase footfall. Integrating stations near such amenities enhances daily usability and multimodal benefits.
- **Bus Stops: 15%.** Maintaining access to current stops supports network continuity and offers potential integration with future express routes.

- **Slope: 10%.** A flat area is necessary for setting up a bus stand to ensure stability and safety for buses during parking, boarding, and alighting. It also facilitates easy construction of infrastructure like shelters, signage, and accessible pathways for passengers.
- **Walk-cycleways: 10%.** Proximity to walkways and cycleways is crucial for a bus stand to ensure safe and convenient access for pedestrians and cyclists. It promotes sustainable transport integration and encourages more people to use public transit.
- **Flood: Exclusion (Negative).** Areas prone to flooding are unsuitable due to high infrastructure maintenance costs and public safety concerns. This layer was used negatively, subtracting from the overall score.

Raster Calculation:

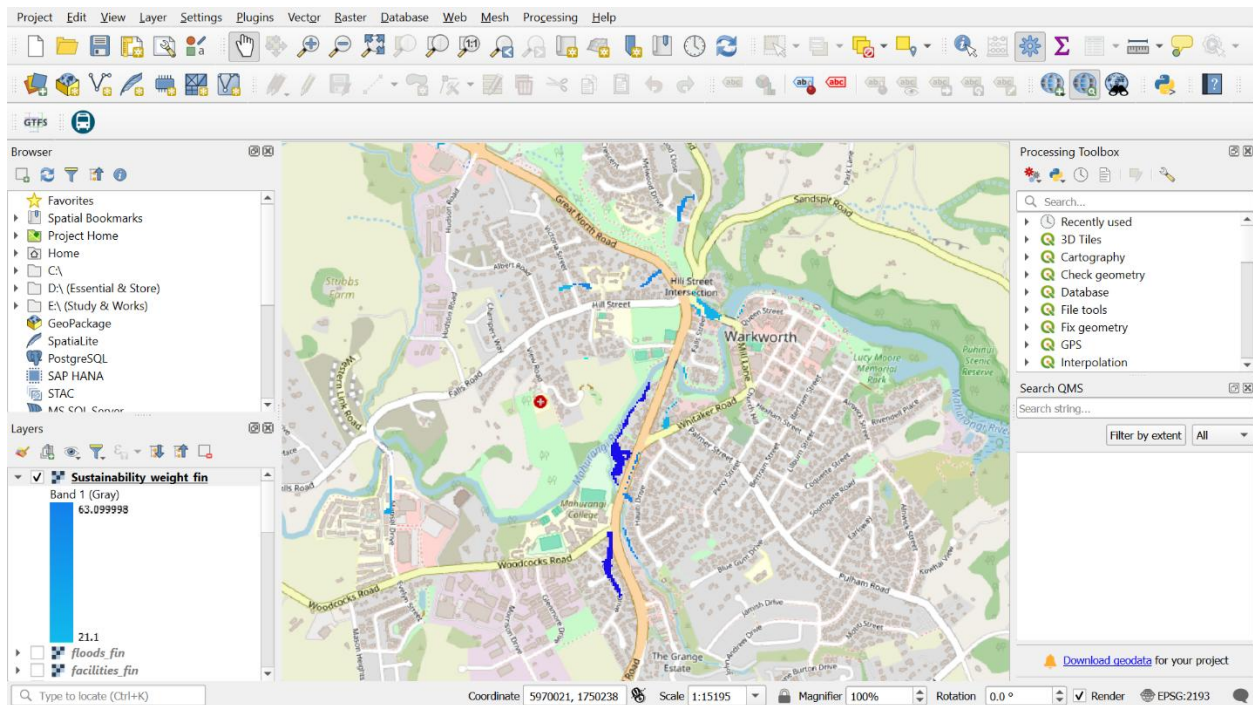
- Raster => Raster Calculator
- Expression: $(\text{"roads_fin@1"} * 1.5 + \text{"population_fin@1"} * 2 + \text{"landuse_fin@1"} * 1.5 + \text{"facilities_fin@1"} * 1.5 + \text{"stops_fin@1"} * 1.5 + \text{"slope_under5@1"} * 1 + \text{"walkways_fin@1"} * 1) - (\text{"floods_fin@1"} * 1.5)$



Raster Calculation

- Symbology => Render type: Singleband Pseudocolor. Color Ramp: Sky Blue to Blue

Following map illustrates the output from raster calculation including the weighted factors and excluding the environmental constraint:



Map 6: Raster calculation output

Geo-visualisation

Geo-visualisation is not only about making maps, it's about transforming analytical results into meaningful spatial narratives. In this project, visualisation played a key role in identifying optimal locations for a new bus station in Warkworth by layering and symbolising complex spatial data into an interpretable and actionable format.

Effective visualisation supports transparency, guides public understanding, and strengthens policy decisions (MacEachren & Kraak, 2001). It also helps to reduce the cognitive load of interpreting multi-criteria analysis outputs by providing a clear visual hierarchy.

Map generalisation techniques were applied where necessary, especially in vector layers such as walk/cycleways and road networks, to simplify geometry and improve readability. For example, minor lanes and alleyways were removed in zoomed-out views to emphasise arterial roads. Simplified symbology was applied using QGIS's style manager to maintain consistency across maps.

While all maps were designed to be visually appealing, efforts were made to avoid manipulative design choices. Suitability weights were applied based on documented planning priorities and academic literature, and colour ramps were linear rather than skewed to emphasise extreme values. The aim was to inform decision-making rather than to steer it toward a pre-conceived outcome, reflecting cartographic ethics discussed by MacEachren (2004).

Software and Tools

All maps were created using QGIS 3.36. Plugins and tools used:

- Raster Calculator: for weighted overlay modelling.
- Processing Toolbox: for rasterizing and buffering.
- GTFS Go Plugin: for loading Bus Stop and Routes dataset.
- QuickMapServices Plugin: for adding OpenStreetMap base layer.
- Color ramp and classification tools: for symbolising raster layers with meaningful gradations.

Visualisation Methodology

The core method for spatial visualisation was the creation of thematic raster maps representing individual suitability factors: slope, flood risk, land use, population density, accessibility to roads, proximity to facilities, and distance to existing bus stops. Each of these was visualised using a continuous or classified colour ramp to reflect relative suitability. These individual raster layers were then combined using a weighted overlay to produce a final suitability map, which was visualised using a multi-colour gradient scale ranging from sky blue (least suitable) to blue (most suitable).

Each map was designed following cartographic best practices to enhance clarity, consistency, and interpretability. The following principles guided design choices:

- Initial layers were classified using a diverging colour ramp (green-yellow-red), where:
 - Green = High suitability
 - Yellow = Moderate suitability
 - Red = Low suitability
- For buffer layers symbology opacity used 50% with different colors for better visibility.
- Final suitability raster was classified using a diverging colour ramp (blue-sky blue) to visually distinguish from other layers and basemap, where:
 - Blue = High suitability
 - Sky Blue = Low suitability
- Colours were chosen to intuitively communicate suitability, in alignment with perceptual expectations.

Reflection on Alternatives

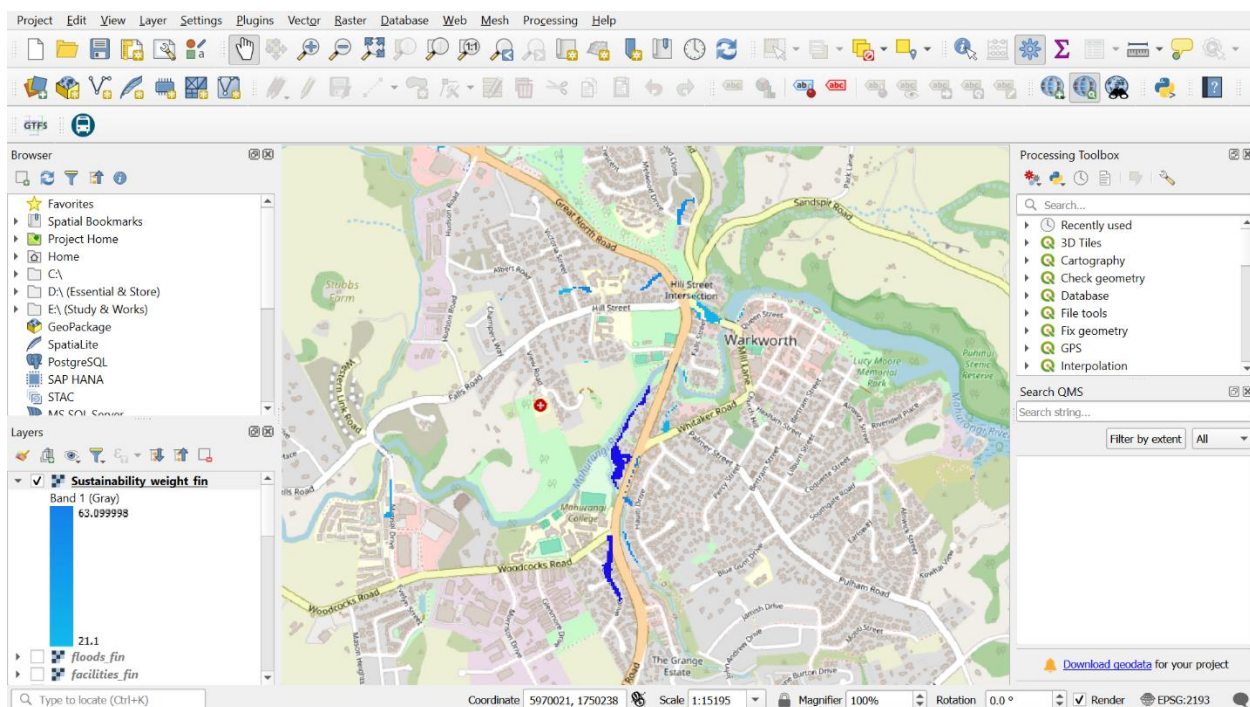
Vector-based MCDA approaches and service area analysis (e.g., using QGIS's network analysis plugin) were considered but rejected due to data limitations and the project's raster-focused methodology. These methods, while useful for network optimisation, are more appropriate when real-time routing data and fine-scale pedestrian network details are available.

3D Terrain Visualisation was considered but not used due to limited need for vertical exaggeration in urban-scale analysis.

Interactive web maps were also explored, but static maps were prioritised as the deliverables required were poster and report.

Final Visualisation

The final sustainability-weighted raster map clearly highlights the most suitable locations of Warkworth, with blue zones indicating optimal bus station sites avoiding green-red colors to distinguish from basemap and clarify visualisation. These areas are located Falls Road, close to the Mahurangi River and west of Woodcocks Road accessible from residential zones, while also avoiding floodplains and steep slopes.



Final Visualisation Map

The visualisation offers a clear, data-driven guide for planners, showing where accessibility, demand, and environmental suitability converge. This output provides a spatially explicit decision-support tool for future infrastructure planning and policy discussions.

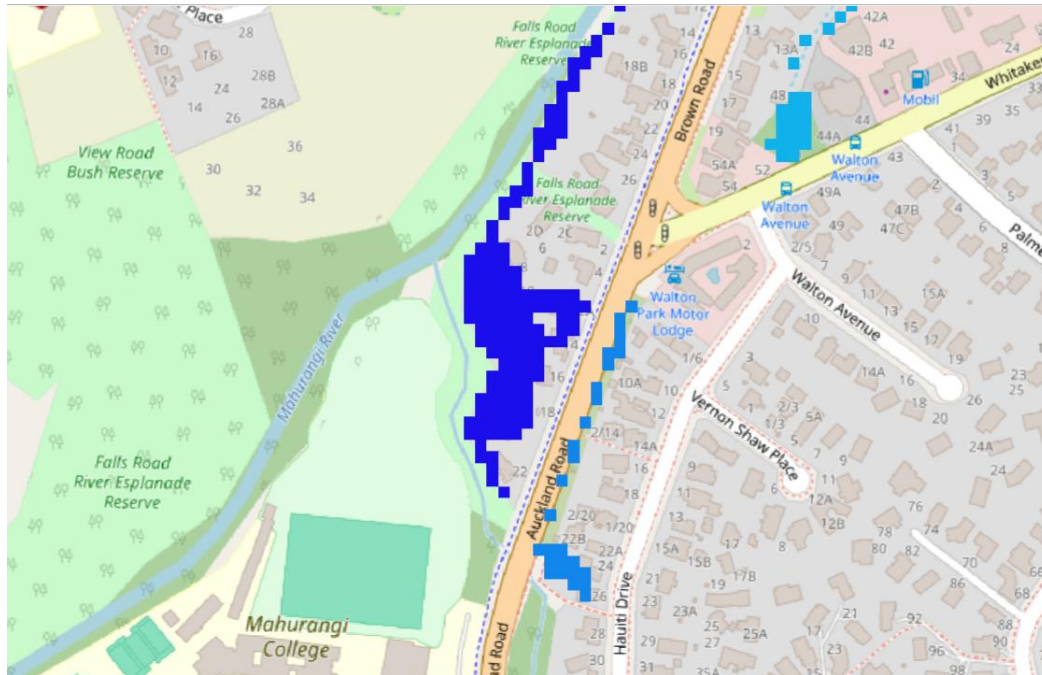
Outcome and Issues

The main question of this project was: Where is the best location in Warkworth for a new bus station when the Northern Expressway (NX1) extends further north? Through a detailed spatial analysis in QGIS, this project identified two key high-suitability areas. These locations were chosen by overlaying multiple rasterised criteria, such as accessibility, population density, environmental safety, and land use compatibility. The final sustainability-weighted raster output visually identifies the best zones in green, representing optimal suitability.

High Sustainability Zones

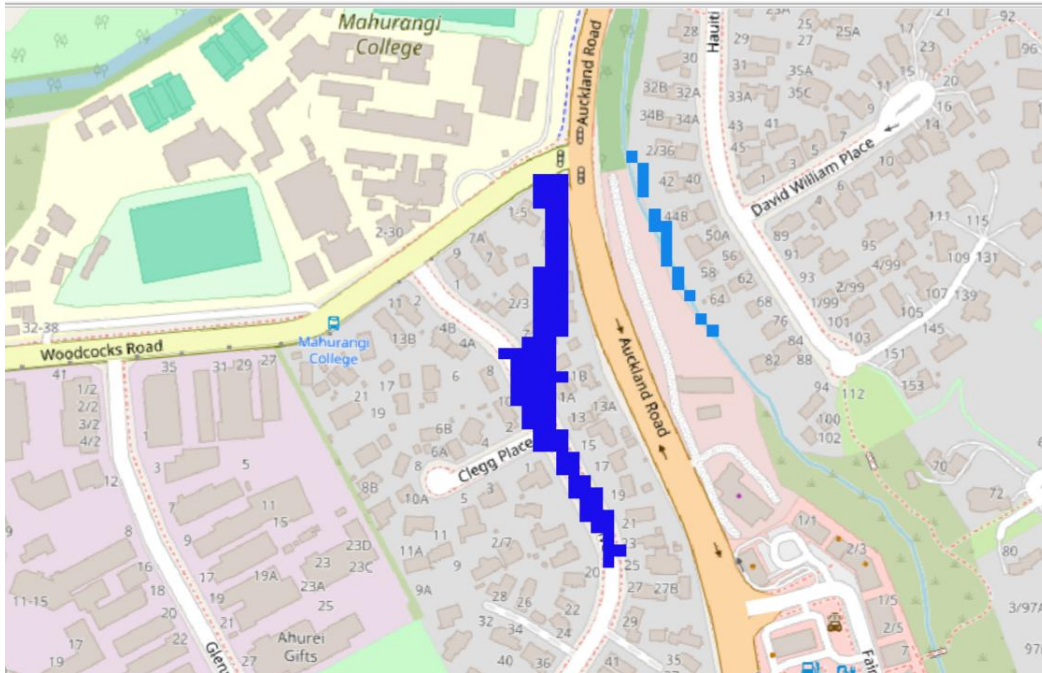
The most suitability zones are concentrated in western Warkworth, particularly around:

- Near Falls Road, close to the Mahurangi River but outside flood-prone areas. This location offers a strategic balance of flat terrain, good road access, and proximity to residential and commercial areas. Around the coordinates -36.402634, 174.658060.



Most Suitable Location 1

- Another promising area was observed A smaller pocket west of Woodcocks Road, offering a connection to arterial transport corridors and adjacent industrial/commercial zoning. Around the coordinates -36.406774, 174.657379.



Most Suitable Location 2

These locations all fall within low-slope zones ($<5^\circ$), outside identified floodplains, and are near existing walk/cycleways, roads, and bus infrastructure. Also, these areas are comparatively closer to State Highway 1 (SH1) that enhances the connectivity to the new bus stop.

Environmental Suitability

Both locations are situated on flat land, with slopes under 5 degrees, ensuring ease of access for people of all abilities. This was confirmed using the DEM-based slope raster. Additionally, both sites fall outside the Auckland Council floodplain zones, avoiding key climate-related hazards. These environmental filters eliminated many otherwise convenient areas, ensuring the chosen sites are resilient and cost-effective for future development.

Analysis Challenges

In the weighted overlay model, accessibility and environmental safety were given the highest weights. This decision was based on literature and practical urban planning priorities. However, balancing multiple factors led to trade-offs. For example, a location close to population centres but near a flood zone was scored lower due to long-term risk.

A few challenges were also encountered:

- Data resolution limitations: GTFS and OSM datasets were rich but may not fully reflect future changes or ground conditions.
- Data reliability: OSM land use data, while accessible and compatible with QGIS, may lack the precision of official zoning datasets.

- Limited qualitative data: Community preferences and behavioural insights were outside the scope of this analysis but would enhance future planning.

Limitations and Future Considerations

Some aspects not included in this project but crucial for real-world decision-making include:

- Economic feasibility and cost of land acquisition,
- Land ownership and availability,
- Stakeholder input and public consultation.

Additionally, future work could integrate transport modelling or simulation tools to predict usage and network performance once the bus station is operational.

The sustainability-weighted raster provided a clear and persuasive outcome. The southeastern and central-eastern zones stand out as the best options for siting a new bus station in Warkworth. These areas are not only well-connected and highly accessible but also avoid environmental risks and are located near key facilities and dense population clusters. This output supports long-term public transport sustainability, safety, and efficiency ensuring Warkworth is well-prepared to grow into a key transport hub in Auckland's northern corridor.

Reflection

This project was a valuable opportunity to apply spatial analysis techniques in a real-world context, with the goal of determining the most suitable location for a future bus station in Warkworth. Overall, the workflow was successful, resulting in a clearly visualised and well-justified final output. By integrating diverse datasets, applying a multi-criteria decision analysis (MCDA) framework, and executing a consistent GIS workflow in QGIS, the project achieved a meaningful and replicable result.

Workflow Effectiveness

The step-by-step spatial workflow was methodically planned and followed a logical structure. From identifying and preparing datasets to reclassifying rasters and performing the weighted overlay analysis, each task contributed directly to answering the project question. The decision to weight slope, flood risk, and accessibility higher than other factors ensured that environmental constraints and public usability were prioritised. The final sustainability-weighted raster map highlighted realistic, appropriate locations that matched the planning goals and practical considerations of Warkworth's urban growth.

The use of QGIS allowed for efficient data processing and cartographic design. The model was flexible enough to accommodate revisions and reruns without compromising the integrity of the results. The visualisation of each step (e.g. slope suitability, floodplain exclusion, and walkability buffers) ensured transparency in decision-making and made it easier to interpret results visually.

Issues Encountered and Solutions

Despite the overall success, several challenges arose. One major issue was the absence of official land parcel or zoning data, which could have offered more precise constraints around land availability and legal feasibility. Instead, the project used OpenStreetMap (OSM) land use data as a substitute, which while effective, had limitations in classification accuracy. Cross-checking OSM layers with aerial basemaps helped mitigate this issue, though a more official land use dataset would improve future accuracy.

Another issue was raster resolution and data compatibility. For example, the DEM slope layer and road network vector data required re-projection and careful alignment. Some raster reclassifications also required trial and error to determine the appropriate breakpoints and suitability scales. These challenges were addressed through consistent CRS management (EPSG:2193), raster alignment tools, and test iterations.

Additionally, the weighted overlay method, while useful, involved subjective decisions when assigning weights. Academic references and local transport planning principles were used to justify these choices, but in future, stakeholder input or sensitivity analysis could be incorporated to further validate weighting assumptions.

What Would I Do Differently?

If repeating the project, I would aim to incorporate zoning and land ownership data from Auckland Council to assess the feasibility of land acquisition or redevelopment more accurately. I would also explore network analysis methods using road travel time rather than just Euclidean distance buffers, which would offer a more realistic understanding of accessibility. Moreover, a suitability threshold analysis could be applied to identify how much land area falls into each suitability class, adding quantitative strength to location selection.

In summary, this project demonstrated a strong and replicable GIS-based approach to solving a real-world planning problem. While minor limitations were present, the methodology proved effective in producing clear, justified outcomes. The lessons learned will help refine future spatial analyses, particularly in balancing data availability with modelling accuracy and in integrating transport equity into infrastructure planning.

Conclusion

This project aimed to identify the most suitable location for a new bus station in Warkworth using spatial analysis tools and a sustainability-focused approach. With the expected extension of the Northern Expressway bus service, Warkworth is becoming a key area for future transport development. The analysis considered a range of geographic and urban factors such as slope, flood risk, population distribution, land use, road access, and proximity to existing facilities and transport infrastructure.

Using a multi-criteria decision analysis (MCDA) model within QGIS, all datasets were reclassified, weighted, and combined into a final sustainability suitability raster. This allowed for the identification of areas that met all key requirements such as accessibility, safety, environmental resilience, and community demand. The most suitable locations identified through this analysis are near Falls Road, close to the Mahurangi River, and a smaller high-suitability area west of Woodcocks Road. These locations are not only accessible and relatively flat but also positioned near population centres and major transport corridors.

The workflow followed a logical structure from data gathering to visualisation, supported by cartographic best practices and spatial reasoning. The result is a clear, evidence-based recommendation for future infrastructure planning in Warkworth. Despite some data limitations, the project achieved its objective and demonstrated the value of GIS in solving real-world transport planning problems. With further community and council input, these findings can contribute meaningfully to future urban development strategies for Warkworth.

References

- Auckland Council. (2020). Auckland Unitary Plan: Operative in Part. <https://unitaryplan.aucklandcouncil.govt.nz/>
- Auckland Council. (2017). Auckland's response to flood risk under climate change: Technical report 2017/019. <https://knowledgeauckland.org.nz/media/1144/tr2017-019-aucklands-response-to-flood-risk-under-climate-change.pdf>
- Auckland Transport. (2019). Transport Design Manual – Streets and Roads. <https://at.govt.nz>
- Auckland Transport. (2021). Hibiscus Coast Busway Station Upgrade Project. <https://at.govt.nz/projects-roadworks/hibiscus-coast-station-upgrade/>
- Bhaskar, A., & Chung, E. (2013). Fundamental understanding on the use of Bluetooth scanner as a complementary transport data. *Transportation Research Part C: Emerging Technologies*, 37, 42–72. <https://doi.org/10.1016/j.trc.2013.09.009>
- Currie, G., & Delbosc, A. (2011). Exploring public transport usage trends in an ageing population. *Transportation*, 38(4), 763–772. <https://doi.org/10.1007/s11116-010-9328-9>
- Curtis, C., & Scheurer, J. (2016). *Planning for Public Transport Accessibility: An International Sourcebook*. Routledge.
- Farahani, R. Z., Miandoabchi, E., Szeto, W. Y., & Rashidi, H. (2013). A review of urban transportation network design problems. *European Journal of Operational Research*, 229(2), 281–302. <https://doi.org/10.1016/j.ejor.2013.01.001>
- Krieger, A., & Higgins, W. (2002). *Urban Design*. University of Minnesota Press.

- MacEachren, A. M., & Kraak, M. J. (2001). Research challenges in geovisualization. *Cartography and Geographic Information Science*, 28(1), 3–12.
<https://doi.org/10.1559/152304001782173970>
- MacEachren, A. M. (2004). *How maps work: Representation, visualization, and design* (2nd ed.). Guilford Press.
- Malczewski, J. (2006). GIS-based multicriteria decision analysis: a survey of the literature. *International Journal of Geographical Information Science*, 20(7), 703–726.
<https://doi.org/10.1080/13658810600661508>
- Ministry for the Environment. (2020). *Preparing for Coastal Change: Guidance for Local Government*. <https://environment.govt.nz>
- Ministry of Transport. (2021). *Keeping cities moving: The role of public transport*. New Zealand Government. <https://www.transport.govt.nz/assets/Uploads/Report/Keeping-Cities-Moving-The-Role-of-Public-Transport.pdf>
- Nasiri, M., Ghorashi, S., & Rezaei-Moghaddam, K. (2018). GIS-based multi-criteria analysis for bus station site selection: A case study. *Transport Policy*, 67, 48–58.
<https://doi.org/10.1016/j.tranpol.2018.04.007>
- New Zealand Transport Agency (NZTA). (2023). *Data Services and Open Data Portal*. <https://www.nzta.govt.nz/data/>